

Lecture 07 - Sampling Distributions and Confidence Intervals

Lecture 7 moves from describing data and probability models to statistical inference. The central question is practical: how can we learn something reliable about a population when we only observe a sample?

In business, finance, manufacturing, marketing, and risk management, we often cannot observe everything we care about. We may not be able to test every product, survey every customer, observe every future return, or measure every transaction. Instead, we take a sample, compute an estimate, and use that estimate to say something cautious about a larger population or process.

The lecture has two parts. The first part introduces sampling: why it is needed, what makes a sample useful, and why different samples give different estimates. The second part introduces confidence intervals: a way to report not only an estimate, but also the precision of that estimate.

1. Why Sampling Is Needed

The motivating example in the slides is a manufacturer of climbing ropes. The company needs to certify that ropes satisfy strength requirements. But the only way to measure the true breaking strength of a rope is to pull it until it breaks. Testing every rope would destroy the entire production.

This is the practical reason for sampling. We test a subset of ropes, measure their strength, and use the sample to infer something about all the ropes produced.

The same logic appears in many domains:

1. A retailer surveys a sample of customers to estimate average satisfaction.
2. A bank studies a sample of loans to estimate default risk.
3. An analyst uses a sample of historical returns to estimate expected return or volatility.
4. A platform uses a sample of users to evaluate a recommendation system.
5. A quality-control team tests a sample of manufactured pieces to assess production quality.

In each case, the sample is not the final object of interest. The sample is the evidence. The population or process is the target.

2. Sampling and Statistical Inference

Sampling is the act of selecting or measuring a subset of observations from a larger population. Statistical inference is the act of using that subset to say something about the larger population.

The distinction matters:

1. **Sampling** is the data collection step.
2. **Inference** is the reasoning step.

A good calculation cannot rescue a bad sample. If the selected observations are not representative of the population, then the estimate may be biased even if the formula is correct. For example, a customer survey collected only from very satisfied customers can produce a precise but misleading estimate of average satisfaction.

This is why sampling design matters. Statistical precision is not only about sample size. It is also about whether the sample is relevant to the question.

3. Database Sampling and Physical Data Collection

Sometimes the data already exist. A company may have a database of users, orders, transactions, or product reviews. In that setting, sampling may mean selecting a subset of rows from an existing dataset.

In other settings, the data do not yet exist. We may need to collect them physically: testing ropes until they break, asking people to try a product, measuring delivery times, or interviewing customers.

The distinction matters because data collection can introduce bias. If a survey is conducted only in one location, at one time of day, or among people who volunteered to respond, then the sample may not represent the target population.

The quality of the sample determines what kind of conclusion is legitimate.

4. Types of Sampling

The slides discuss several common sampling approaches.

Method	Definition	Benefit	Potential bias
Random sampling	Each observation has a chance to be selected.	Objective and often cost efficient.	Small groups may be missed by chance.
Systematic sampling	Observations are selected at regular intervals.	Simple and can be representative.	Cycles in the data can make the sample unrepresentative.
Quota sampling	The sample is chosen to match subgroup proportions.	Ensures selected groups are represented.	Depends on which quotas are chosen and how units are selected within quotas.

Random sampling is conceptually clean because it gives a probabilistic basis for inference. Systematic sampling can be useful, but it can fail if the interval matches a pattern in the data. Quota sampling can improve representation, but it involves judgment about which groups matter.

The practical lesson is that sampling is not just an administrative detail. It shapes the meaning of the estimate.

5. Sampling Fluctuation

Suppose we take one sample and compute the sample mean. Then suppose we take another sample from the same population and compute another sample mean. The two sample means will usually differ.

This difference is called sampling fluctuation. It is not a mistake. It is the natural consequence of using partial information.

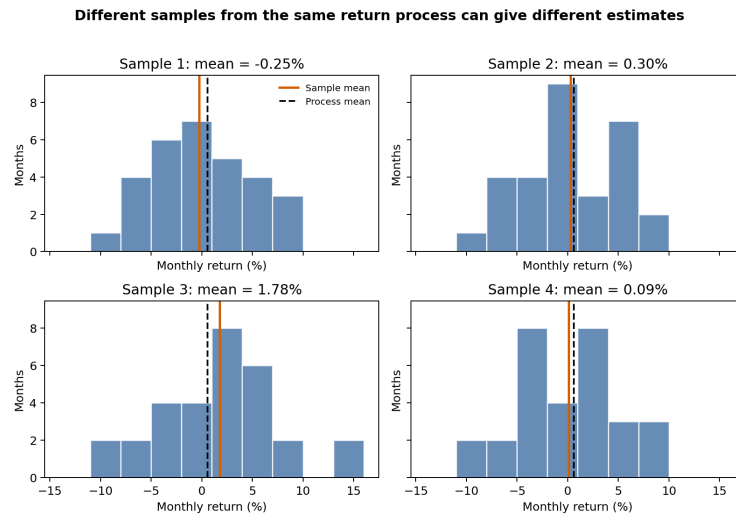


Figure 1. Different samples drawn from the same process can produce different sample means.

The figure shows several samples drawn from the same underlying return process. Each sample has its own mean. Some sample means are below the process mean, and some are above it. The variability across sample means is the object we need to understand in order to discuss precision.

6. Parameter, Estimator, Estimate, and Estimation Error

Lecture 3 introduced the distinction between parameters, estimators, and estimates. Lecture 7 uses that distinction again.

The parameter is the unknown population quantity we care about. Examples include the true average rope strength, the true average customer waiting time, or the true expected return of a strategy.

The estimator is the rule used to estimate that parameter. For a population mean, the usual estimator is the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

The estimate is the number obtained when the estimator is applied to a particular sample.

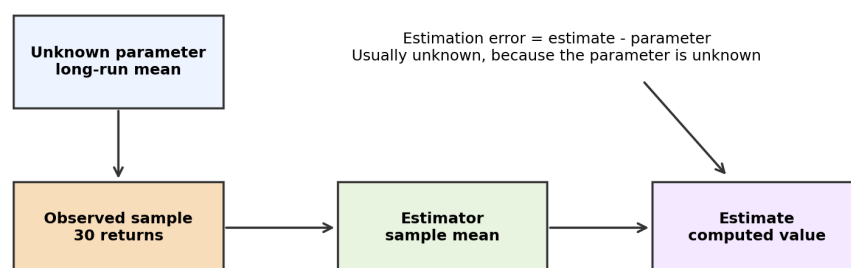


Figure 2. The estimator is a rule; the estimate is the number produced by that rule on one sample.

The estimation error is

$$\text{estimation error} = \text{estimate} - \text{parameter}.$$

In real applications, the estimation error is usually unknown because the parameter is unknown. Confidence intervals are useful because they provide a way to communicate how large this error might plausibly be.

7. The Sampling Distribution

If we repeatedly draw samples of the same size from the same population and compute the sample mean each time, the sample means themselves form a distribution. This distribution is called the sampling distribution of the sample mean.

The sampling distribution is not the distribution of individual observations. It is the distribution of an estimator across repeated samples.

This distinction is essential:

1. The standard deviation describes variability of individual observations.
2. The standard error describes variability of an estimate across samples.

If individual returns are volatile, the standard deviation is large. If an estimated average return would vary a lot from sample to sample, the standard error is large.

8. Standard Error of the Sample Mean

If the population standard deviation is σ and the sample size is n , the standard error of the sample mean is

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

This formula contains two important ideas.

First, greater individual variability increases uncertainty in the sample mean. If the population standard deviation is large, sample means vary more from sample to sample.

Second, larger samples reduce uncertainty. The denominator is $\sqrt{n}$, so the standard error decreases as the sample size increases.

The square-root relationship is important. Multiplying the sample size by 4 divides the standard error by 2. To reduce the standard error by a factor of 10, the sample size must be multiplied by 100.

In practice, σ is often unknown. Then we estimate the standard error using the sample standard deviation:

$$\widehat{SE}(\bar{X}) = \frac{s}{\sqrt{n}}.$$

9. Sampling Distributions Become Tighter With Larger Samples

As sample size increases, the distribution of sample means becomes more concentrated around the true mean.

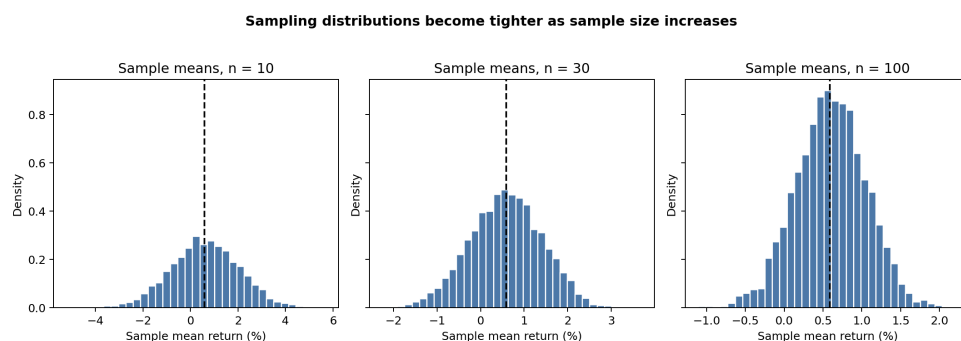


Figure 3. Larger samples produce sampling distributions with smaller standard errors.

The figure shows sampling distributions for different sample sizes. When n is small, sample means vary widely. When n is larger, sample means are more tightly concentrated. This is the statistical meaning of “more precise.”

A larger sample does not guarantee that the estimate equals the parameter. It does, however, reduce the typical sampling fluctuation when the sampling process is appropriate.

10. The Central Limit Theorem

The Central Limit Theorem explains why normal approximations appear so often in inference.

If the sample size is large enough, the sampling distribution of the sample mean is approximately normal under broad conditions, even when the original data are not normally distributed.

This does not say that the raw data become normal. It says that the distribution of the sample mean becomes approximately normal.

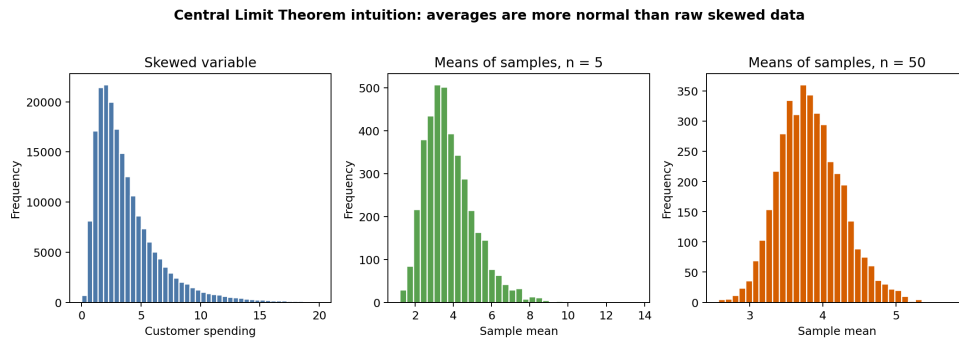


Figure 4. A skewed variable can produce sample means that become more symmetric and normal-looking as sample size increases.

The theorem is one reason confidence intervals for means often use normal critical values such as 1.645, 1.96, and 2.58. The approximation works best when samples are large, observations are reasonably independent, and the underlying distribution is not dominated by extreme values.

11. Netflix Viewing Example: Standard Error

The slides use a simple example based on weekly Netflix viewing hours among European families.

Suppose the population mean is

$$\mu = 6$$

hours per week, the population standard deviation is

$$\sigma = 1.2$$

hours, and the sample size is

$$n = 100.$$

If we repeatedly take samples of 100 households and compute the sample mean each time, the average of those sample means should be close to 6 hours. But individual sample means will fluctuate.

The standard error is

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{1.2}{\sqrt{100}} = \frac{1.2}{10} = 0.12.$$

So the sample mean viewing time typically fluctuates by about 0.12 hours around the population mean across repeated samples of size 100.

This is much smaller than the population standard deviation of 1.2 hours. The standard deviation describes differences among households. The standard error describes uncertainty in the sample average.

12. From Sampling Distribution to Confidence Interval

The next question is how to turn standard error into an interval.

If the sampling distribution of the sample mean is approximately normal, then central regions of the normal curve tell us how far sample means usually fall from the center.

For the standard normal distribution:

1. about 90% of values lie between -1.645 and 1.645 ;
2. about 95% of values lie between -1.96 and 1.96 ;
3. about 99% of values lie between -2.58 and 2.58 .

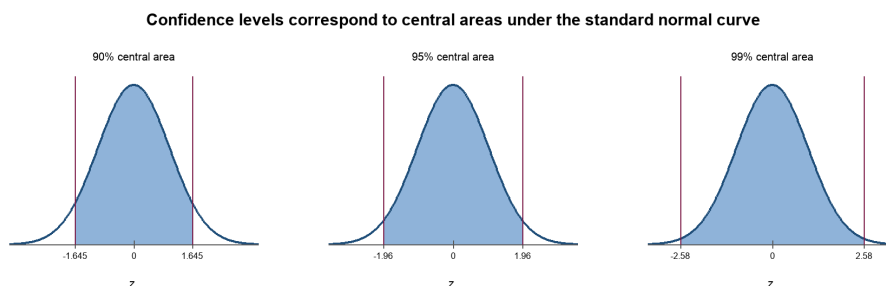


Figure 5. Higher confidence levels require wider central regions under the standard normal curve.

The interval width depends on the critical value. A higher confidence level uses a larger critical value and therefore gives a wider interval. This is the precision-confidence tradeoff: a narrow interval is more precise, but less likely to capture the target under repeated sampling; a wider interval is less precise, but corresponds to a higher confidence level.

13. Confidence Intervals for Sample Means

The basic confidence interval structure is

$$\text{estimate} \pm \text{critical value} \times \text{standard error}.$$

For a mean, this becomes

$$\bar{X} \pm z^* \frac{s}{\sqrt{n}}$$

when the standard error is estimated with the sample standard deviation.

In the simplified setting used in the slides, if the population mean and standard deviation are treated as known, a central interval for the sampling distribution of the sample mean is

$$\mu \pm z^* \frac{\sigma}{\sqrt{n}}.$$

In practice, however, we usually do not know μ . We observe a sample mean $\bar{X}$, estimate the standard error, and report an interval around the estimate as a plausible range for the unknown population mean.

14. Netflix Confidence Interval Example

Return to the Netflix viewing example:

$$\mu = 6, \quad \sigma = 1.2, \quad n = 100.$$

For a 95% central interval, the critical value is 1.96. The standard error is 0.12. The margin is

$$1.96 \times 0.12 = 0.2352.$$

The interval is therefore

$$6 \pm 0.2352 = [5.7648, 6.2352].$$

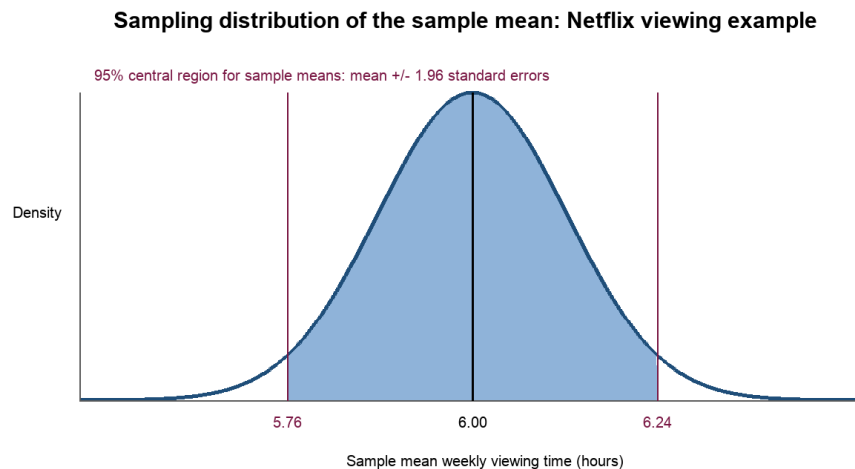


Figure 6. For samples of size 100, about 95% of sample means fall between 5.76 and 6.24 hours in this model.

The interpretation in this known-parameter example is: if we repeatedly take samples of 100 households, about 95% of the sample means will fall between 5.76 and 6.24 hours.

In a real estimation problem, the parameter is unknown, so the interval is usually reported around the observed sample mean and interpreted as a plausible range for the unknown population mean under the assumptions of the method.

15. What Confidence Intervals Do Not Mean

A confidence interval for a mean is not a range that contains 95% of individual observations.

For example, if a 95% confidence interval for average waiting time is [3.6, 4.4] minutes, it does not mean that 95% of customers waited between 3.6 and 4.4 minutes. Individual waiting times may be much more dispersed.

The interval describes uncertainty about the average, not variability of individual observations.

It is also not correct to say, after computing one interval, that there is a 95% probability that the fixed true mean is inside that specific interval. In the frequentist interpretation, the method has 95% coverage across repeated samples. In applied work, a practical careful statement is:

Based on this sample and the assumptions of the method, the interval gives a plausible range for the unknown population mean.

16. Precision, Confidence, and Caution

The width of a confidence interval is driven by three forces:

1. the variability of the data;
2. the sample size;
3. the chosen confidence level.

More variable data produce wider intervals. Larger samples produce narrower intervals. Higher confidence levels produce wider intervals.

A narrow interval means statistical precision. It does not automatically mean economic importance. For example, an estimated average return can be very precisely near zero and still be economically uninteresting. Conversely, a wide interval can include economically important positive and negative possibilities, warning us not to overclaim.

The final habit of the lecture is simple:

Do not report only the estimate. Report how precise it is.

This habit prepares the course for hypothesis testing and regression inference. Both build on the same logic: an estimate is only part of the story; uncertainty around the estimate is just as important.